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EDUCATION

Farmingdale State College

Farmingdale, NY

Bachelor of Science in Computer Science, Coursework in Applied Mathematics

May 2026

- Cumulative GPA: 3.40
- Relevant Coursework: SQL, Python, Multivariable Calculus, Linear Algebra, Statistics

PROJECTS

Tesla Stock Reversal Prediction (ML, Tableau)

Remote

Self-Initiated Research Project

June 2025

- Engineered 7 ML features from Tesla stock data via SQL preprocessing and Python feature derivation (1st/2nd derivatives).
- Trained a scikit-learn RandomForest on 339-time steps with 5-fold TimeSeriesSplit for a 0.75 F1-score on reversal detection.
- Integrated an end-to-end ETL pipeline (SQL-Python-Tableau) with reversal markers across 16 months of Tesla stock-data.

HR Diagram-Based Stellar Classification Analysis (ML)

Remote

Self-Initiated Research Project

May 2025

- Processed 240 stellar records and engineered 5 features (Pandas + NumPy), including log-scaled temperature and luminosity.
- Trained a linear regression model to view spectral patterns and show HR diagram groupings (main sequence, giants, dwarfs).
- Applied IQR-based outlier filtering and visualized raw vs log-scaled relationships in 2 plots using Matplotlib and Seaborn.

3D Level-Based Survival Game

Remote

Non-Commercial Developer

July 2024 – Present

- Developed 3D maps and 30+ assets in Unreal Engine/Blender, with survival mechanics inspired by Call of Duty: Zombies.
- Scripted 1000+ Blueprint/C++ functions to enhance animation, 3D-object modeling and optimized performance.
- Engineered 3 AI-enemy types with pathfinding, perception and combat-state logic for responsive gameplay.

PROFESSIONAL EXPERIENCE

Power Device Corporation

Bohemia, NY

Semiconductor Test Technician

Jan 2025 – Present

- Conducted thermal stress and reliability testing on semiconductors, collecting data to assess compliance with thresholds.
- Logged 200+ device results daily, using data analysis and statistical quality control to detect performance trends and failures.
- Collaborated with engineers to analyze test outcomes and guide process adjustments to improve production reliability.

LANRover Network Services

Wantagh, NY

Information Technology Intern

Jul 2024 – Oct 2024

- Collected and maintained hardware inventory data across 3+ school districts, improving accuracy of asset documentation.
- Standardized device records and labeling practices, supporting 10+ IT staff in tracking and deployment planning.
- Structured asset data to ensure consistency across districts and enabled efficient lookup during troubleshooting.

LEADERSHIP

Homeland Security Investigations Explorer Post 601

New York, NY

Program Manager

Sep 2024 – Present

- Collaborated with advisors and peers in developing the HSI Explorer Post 601 website, driving online engagement by 25%.
- Coordinated interactive mock calls, suspect confrontations, arrest warrant simulations, and witness interview exercises.
- Facilitated 30+ weekly sessions with federal agents to provide experience in law enforcement and criminal justice.

SKILLS

Mathematics: Calculus (I-III), Linear Algebra, Probability & Statistics, Optimization, Game Physics

Data Science: Machine Learning, Feature Engineering, Exploratory Data Analysis, ETL Pipelines, Time Series Forecasting

Tools/Tech: Python (Pandas, NumPy, Matplotlib, Seaborn, Scikit-Learn), SQL Server Management Studio, C++, Java, GitHub, Tableau, Jupyter Notebooks, Firebase Studio, Unreal Engine, Adobe After Effects